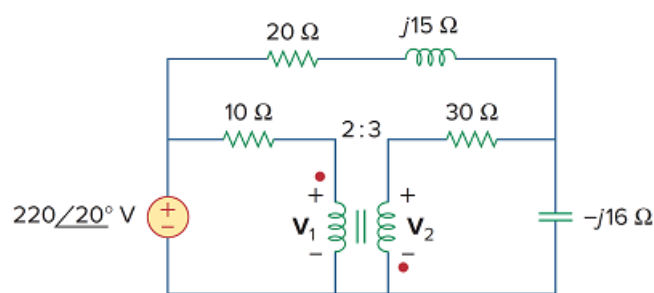


Problem

Obtain V_1 and V_2 in the circuit of Fig. 13.58 using PSpice.

Figure 13.58



Step-by-step solution

Step 1 of 9

Refer to Figure 13.58 from the textbook.

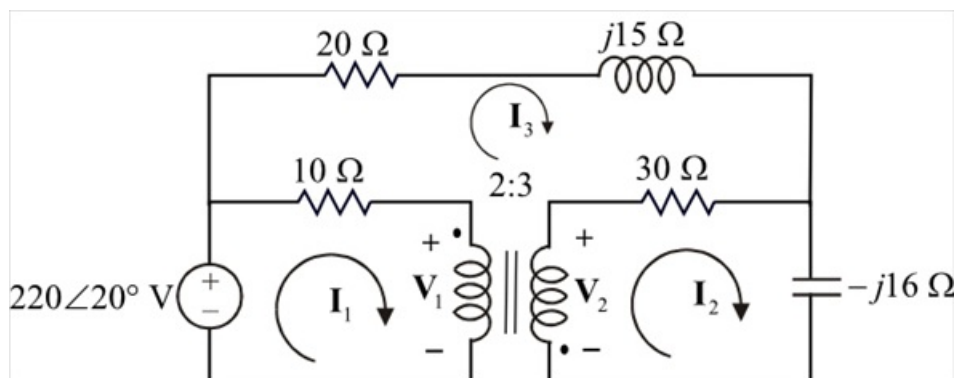


Figure 1

Step 2 of 9

Apply Kirchhoff's voltage for loop I_1 .

$$-220\angle 20^\circ + 10(I_1 - I_3) + V_1 = 0$$

$$10I_1 - 10I_3 + V_1 = 206.73 + j75.24 \quad \dots\dots (1)$$

Apply Kirchhoff's voltage for loop I_2 .

$$-V_2 + 30(I_2 - I_3) - j16I_2 = 0$$

$$(30 - j16)I_2 - 30I_3 - V_2 = 0 \quad \dots\dots (2)$$

Apply Kirchhoff's voltage for loop I_3 .

$$-V_1 + 10(I_3 - I_1) + (20 + j15)I_3 + 30(I_3 - I_2) + V_2 = 0$$

$$-10I_1 - 30I_2 + (60 + j15)I_3 - V_1 + V_2 = 0 \quad \dots\dots (3)$$

Step 3 of 9

Write the relation between voltages and turns ratio.

$$\frac{V_2}{V_1} = -\frac{N_2}{N_1}$$

$$\frac{V_2}{V_1} = -\frac{3}{2}$$

$$2V_2 = -3V_1$$

$$3V_1 + 2V_2 = 0 \dots\dots (4)$$

Write the relation between currents and turns ratio.

$$\frac{I_2}{I_1} = -\frac{N_1}{N_2}$$

Subtract numerator and denominator by I_3 .

$$\frac{I_2 - I_3}{I_1 - I_3} = -\frac{2}{3}$$

$$3I_2 - 3I_3 = -2I_1 + 2I_3$$

$$2I_1 + 3I_2 - 5I_3 = 0 \dots\dots (5)$$

Step 4 of 9

Write the equations (1), (2), (3), (4) and (5) in matrix form.

$$\begin{bmatrix} 10 & 0 & -10 & 1 & 0 \\ 0 & 30 - j16 & -30 & 0 & -1 \\ -10 & -30 & 60 + j15 & -1 & 1 \\ 0 & 0 & 0 & 3 & 2 \\ 2 & 3 & -5 & 0 & 0 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 206.73 + j75.24 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Write the MATLAB code to solve the matrix equation.

```
>> a=[10 0 -10 1 0;0 30-j*16 -30 0 -1;-10 -30 60+j*15 -1 1;0 0 0 3 2;2 3 -5 0 0];
```

```
>> b=[206.73+j*75.24;0;0;0;0];
```

```
>> c=inv(a)*b
```

```
c = 1.0e+02 * 0.1930 + 0.0064i 0.0515 - 0.0081i 0.1081 - 0.0023i 1.2182 + 0.6654i -1.8272 - 0.9981i
```

From the MATLAB output,

$$V_1 = 121.82 + j66.54 \text{ V}$$

$$= 138.8 \angle 28.64^\circ \text{ V}$$

$$V_2 = -182.72 - j99.81 \text{ V}$$

$$= 208.2 \angle -151.35^\circ \text{ V}$$

Therefore, the voltages V_1 and V_2 are $138.8 \angle 28.88^\circ \text{ V}$ and $207.5 \angle -151.7^\circ \text{ V}$.

Step 5 of 9

Calculate the voltages V_1 and V_2 using PSpice.

Step 1:

Select Draw/Get new part and type XFRM_NONLINEAR click ok and place the XFRM_NONLINEAR symbol on the schematic. And double click on XFRM_NONLINEAR device and set coupling = 1, L1_TURNS = 200000 and L2_TURNS = 300000.

Step 2:

The given circuit is as shown in Figure 1, after making all connections by using PSpice.

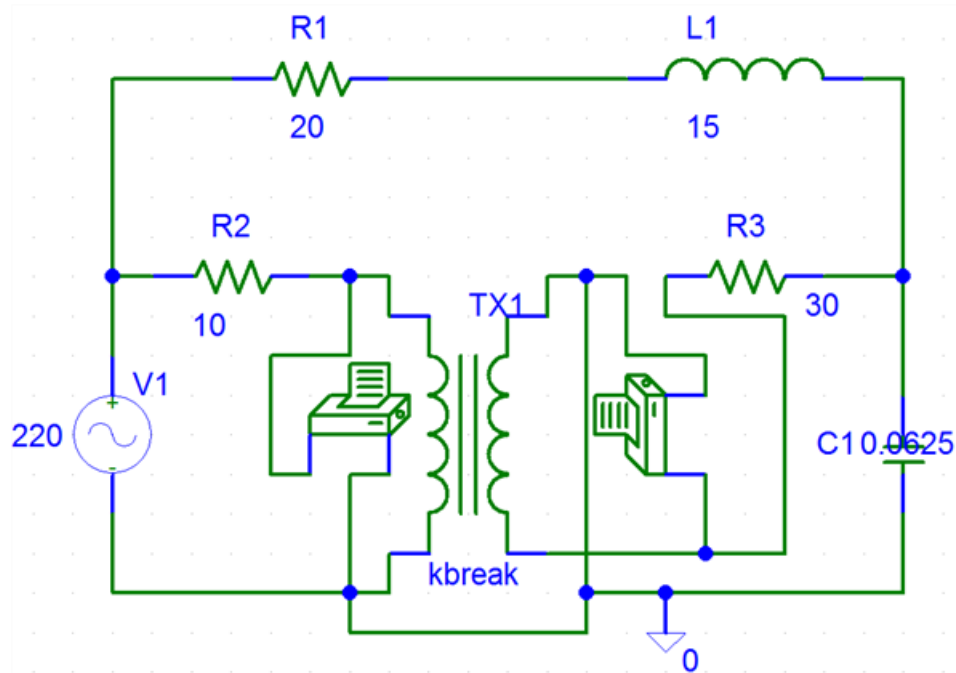


Figure 1

Step 6 of 9

Step 3:

Double click VAC voltage source and set $DC = 0V$, $ACMAG = 220$ and $ACPHASE = 20$ and double click on VPRINT and set $AC = yes$, $MAG = yes$ and $PHASE = yes$.

Step 7 of 9

Step 4:

As usual, we assume $\omega = 1 \text{ rad/s}$ and find the corresponding values of capacitance and inductance of the elements.

$$j15 = j\omega L$$

$$j15 = j(1)L$$

$$L = 15 \text{ H}$$

And,

$$-j16 = \frac{1}{j\omega C}$$

$$-j16 = \frac{1}{j(1)C}$$

$$C = \frac{1}{16}$$

$$C = 0.0625 \text{ F}$$

$$f = 0.1592 \text{ Hz}$$

Go to Analysis/Setup and select AC sweep/linear and enter total points = 1, start frequency = 0.1592Hz and end frequency = 0.1592Hz.

Step 8 of 9**Step 5:**

Select Analysis/Simulate to simulate it. The output file includes.

FREQ VM(\$N_0004,0) VP(\$N_0004,0)

1.592E-01 1.389E+02 2.888E+01

FREQ VM(\$N_0005,0) VP(\$N_0005,0)

1.592E-01 2.075E+02 -1.517E+02

Step 9 of 9

Therefore, the output file shows,

$$\mathbf{V}_1 = 138.8 \angle 28.88^\circ \text{ V}$$

$$\mathbf{V}_2 = 207.5 \angle -151.7^\circ \text{ V}$$

Therefore, the voltages $\mathbf{V}_1$ and $\mathbf{V}_2$ are $138.8 \angle 28.88^\circ \text{ V}$ and $207.5 \angle -151.7^\circ \text{ V}$.

Thus both schematic and theoretical values are same.